EARTH AND PLANETARY SCIENCE LETTERS 16 (1972) 15–22. NORTH-HOLLAND PUBLISHING COMPANY

CHRONOLOGY OF LATE PLIOCENE AND EARLY PLEISTOCENE GLACIAL AND MAGNETIC EVENTS IN SOUTHERN ARGENTINA *

R.J. FLECK
Department of Geology, Ohio State University, Columbus, Ohio 43210, USA

J.H. MERCER
Institute of Polar Studies, Ohio State University, Columbus, Ohio 43210, USA

A.E.M. NAIRN and D.N. PETERSON **
Department of Geology, Case Western Reserve University, Cleveland, Ohio 44106, USA

Received 5 May 1972
Revised version received 21 June 1972

K–Ar and paleomagnetic studies of basalt flows in southern Argentina indicate that the three major normal-polarity events of the Matuyama reversed geomagnetic polarity epoch are recorded in a single cliff section. Till interbedded with the lava flows indicates at least four separate incursions of ice, the earliest prior to 2.06 my ago. Till is present above the basalt flows that record the Réunion event and immediately beneath flows formed during the Olduvai and Jaramillo events.

1. Introduction

In southern Argentina, basalt flows interbedded with till record in one stratigraphic sequence the three major normal-polarity events in the Matuyama reversed geomagnetic polarity epoch. Potassium–argon age determinations and paleomagnetic studies of the flows confirm glaciation of the site more than 2 my ago. The exposure, first described by Feruglio [1], is on Cerro del Fraile (1432 m), a hill situated at lat. 50°33′S, long. 72°40′W, south of Lago Argentino (186 m) (fig. 1). Mercer re-examined the site and collected samples for age determination from the top and bottom flows [2]. He later returned and collected oriented samples from seven flows, from which the duplicate K–Ar analyses and replicate paleomagnetic polarity measurements reported here were made. These studies revise the age assignments originally reported by Mercer [2], which were based on imprecise analyses, but they support his conclusions.

* Contribution No. 230 of the Institute of Polar Studies, The Ohio State University, Columbus, Ohio 43210, USA.

** Present address: Department of Geology, State University College, Fredonia, New York 14063, USA.

Cerro del Fraile is composed of Cretaceous sedimentary deposits unconformably capped at 1020 m above sea level on its western face by a total thickness of about 180 m of late Cenozoic basaltic lava flows and interbedded glacial and fluvial deposits [1]. The stratigraphy of the Cenozoic deposits varies along the exposure, as some units thicken, thin, or pinch out. Fig. 2, after Mercer [2], shows the stratigraphy at Cerro del Fraile. Tills are numbered 1 through 6; units 2 and 4 were thin and discontinuous. Basalt flows are lettered AA through H; flow E was not sampled, and letters A and AA designate the same flow.

2. Analytical techniques

Potassium–argon age determinations were made on at least two separate specimens from each of seven basalt flows at Cerro del Fraile. All analyses were made on whole-rock samples from the oriented hand specimen cored for paleomagnetic study. Rock slices were taken from both ends of each whole-rock sample run for argon, were powdered to less than 100 mesh size, and were split for two separate potassium analyses.

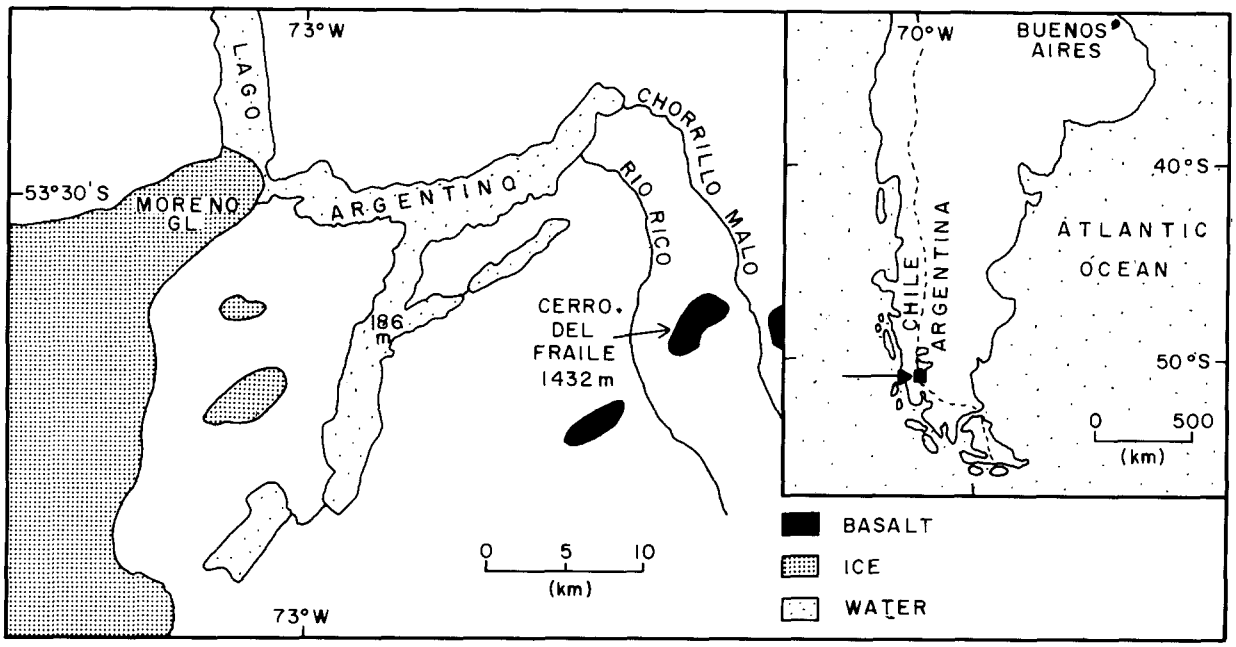


Fig. 1. Map showing location of Cerro del Fraile, Argentina (from Mercer, Science, 164, 823). Copyright by the American Association for the Advancement of Science.

Potassium was determined by flame photometry, using a Zeiss PF-5 flame photometer with propane-air fuel and a cation separation procedure similar to that of Cooper [3]. Argon analyses were made by isotope dilution, using a Nuclide 6-inch radius, 60°-sector mass spectrometer. Samples were fused by induction heating in an evacuated extraction line similar to that described by Evernden and Curtis [4]. Manifold-type tracers with an approximate size of 2×10^{-10} mole ^{38}Ar were used in the analysis. Tracers were calibrated using University of California (Berkeley) and U.S. Geological Survey (Menlo Park) laboratory standard biotites.

Cores were drilled from oriented hand specimens of each of the flows sampled, and the directions of remanent magnetization were measured on a Foster-type, low-speed, spinner magnetometer [5]. The accuracy of the results may be limited by errors in orienting specimens during collection, reorienting the blocks for coring in the laboratory, transferring the orientation marks to the drilled core, or measuring the components of remanence on the magnetometer. The combined error is probably not more than ± 5°, for the individual errors are random and not necessarily cumulative. Magnetic stability was tested and low-coercivity secondary components removed by progressive alternating field demagnetization in peak-field strengths of 77, 150, 300, and 600 oersteds. The mean direction of magnetization was computed for each of these demagnetization stages and the direction for which the value of k (precision parameter) was largest and α_{95} (the semiangle of the cone of 95% confidence) smallest was considered the best estimate of the orientation of the earth's field direction at the time of extrusion. The demagnetization generally changed the directions of magnetization by less than a few degrees. Of the seven flows sampled, three had reversed polarity, three had normal polarity, and one probably represents an intermediate position (fig. 3).

3. Discussion of age determinations

Estimates of the analytical precision of the potassium–argon analyses are shown in table 1 with the "Calculated Age" of the individual specimens. This value, listed as (±), represents an *estimation* of the standard deviation of that analysis and is based on the quality of the individual experiment. This estimate is

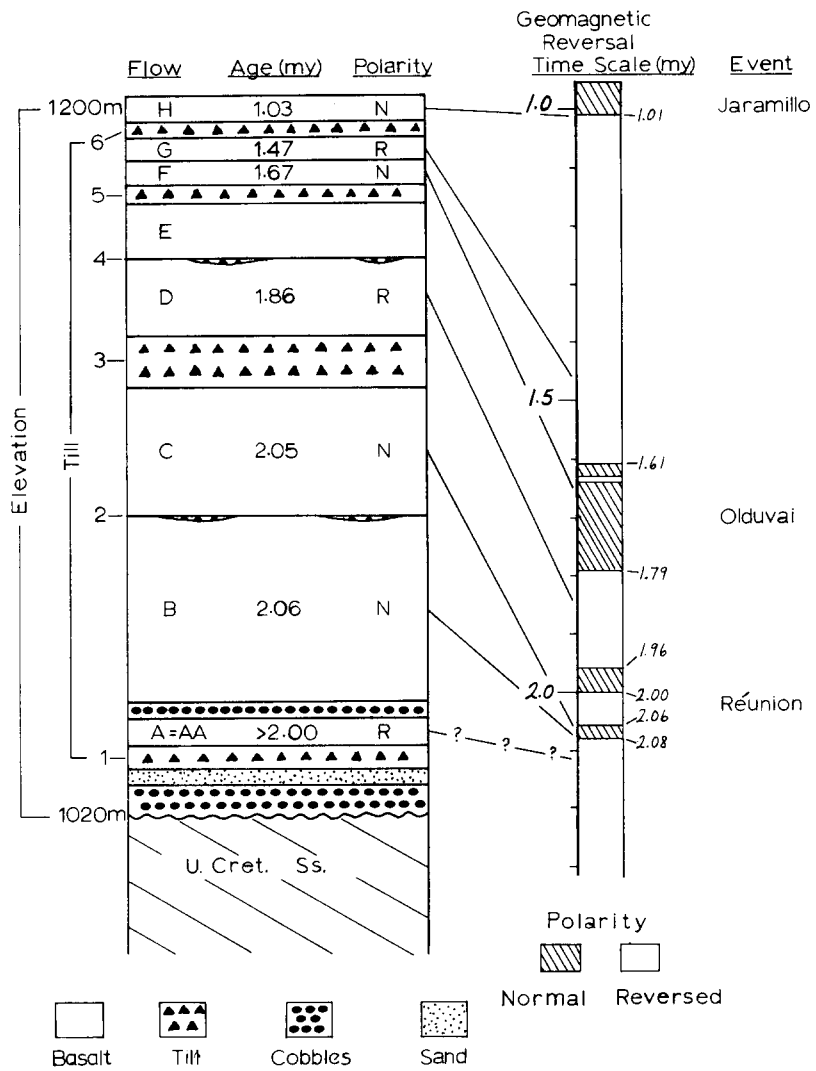


Fig. 2. Stratigraphic section and summary of potassium–argon and paleomagnetic results, Cerro del Fraile.

made in a manner similar to that described by Cox and Dalrymple [6] and includes the precision of tracer calibration and potassium analyses, in addition to terms dependent upon the percentage of radiogenic argon. The factor shown as (±) in the last column of table 1 under "Age" represents the *calculated* standard deviation of all analyses of a given flow unit. Flows A through H were sampled one above the other in the cliff section, but the base of flow A was obscured. In the field flow A was identified as probably being the same unit as flow AA, the lowermost volcanic unit at Cerro del Fraile. Flow AA is exposed at about the same elevation as flow A, at a point 200 m to the south, where it rests on the oldest till in the sequence. As will be noted in table 2, this identification is substantiated by the magnetic data.

All samples analyzed were examined in thin section to eliminate any with detectable alteration or groundmass devitrification. None of those analyzed had either of these aspects and no anomalous results were expected. However, the flow unit from which samples labeled A or AA were collected gave inconsistent results. Age determinations from this flow (table 1) have much larger internal variation than any of the

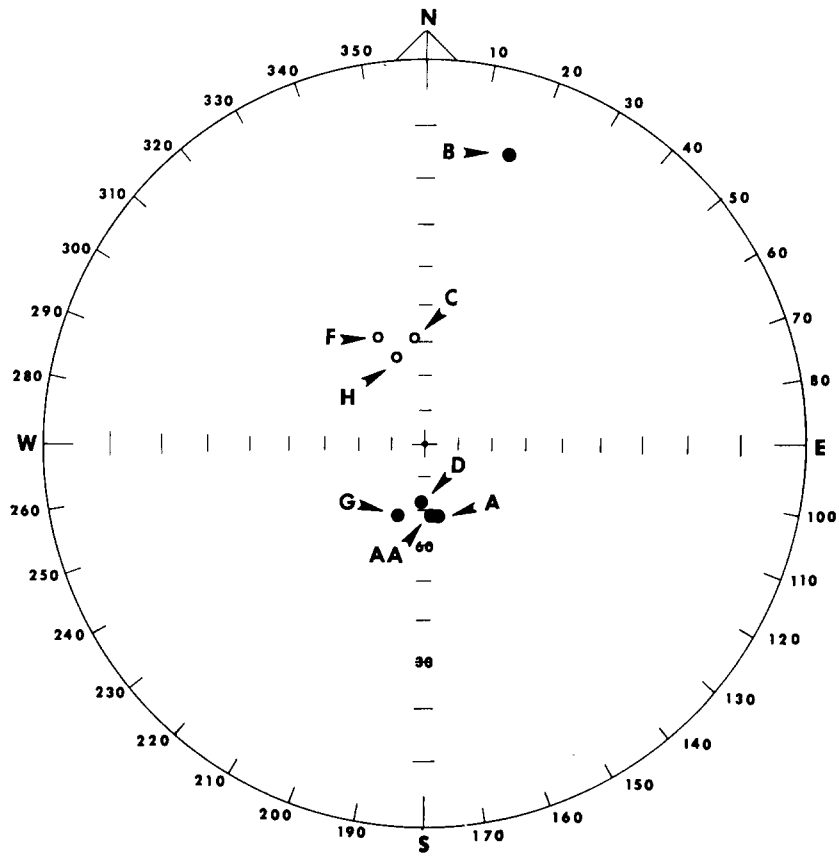


Fig. 3. Mean directions of magnetization. Solid circles, lower hemisphere; open circles, upper hemisphere.

younger flows and are the only analyses that fall out of stratigraphic order. The age determined for unit A alone is not distinguishable from those of flows B and C at the 95% confidence level, and though out of sequence, the difference may be fortuitous. However, the age of the same flow at AA, laterally separated from the cliff section of A through H by 200 m, is statistically different and suggest a loss of radiogenic argon. Except for the lowermost flow, the reproducibility of the age determinations demonstrates the intraflow consistency and indicates that loss or addition of radiogenic argon is improbable. The reliability of the method used to estimate the precision of a single analysis is also indicated by the actual determinations.

4. Discussion of results

Fig. 2 summarizes the paleomagnetic and potassium–argon results for the basalt flows at Cerro del Fraile. Periods of normal polarity at 2.05, 1.67 and 1.03 my are correlated with the Réunion, Olduvai and Jaramillo events, respectively. Flows D and G with reversed polarity separate these events, confirming the presence of at least three normal polarity events in the Matuyama reversed epoch. The intermediate polarity shown for flow B suggests that a reversal may have occurred at about 2.06 my, with the earth's field going from reversed to normal polarity. The 2.06 and 2.05 my ages of flows B and C fall between the two periods of normal polarity assigned by Cox [8] to the earliest event in the Matuyama reversed epoch, here called the Réunion, as suggested by Grommé and Hay [9]. The normal event recorded by flows B and C could conceivably represent either of these two periods, or a third short interval not recognized previously. A recent summary by Dalrymple [10] of radiometrically dated flows of established polarity indicates age ranges of normal flows from 1.95 to 2.01 my and 2.08 to 2.09 my with no intervening values. If the values for

Table 1
Potassium–argon age determinations.

Flow	Sample number	Percent K^+	Moles ^{40}Ar */g ($\times 10^{-12}$)	Percent ^{40}Ar *	Calculated age ** (my)	Age *** (my)
H	34	2.018, 2.018	3.644	15.6	1.01 ± 0.04	
	36	1.849, 1.850	{3.618	27.9}	1.04 ± 0.05	1.03 ± 0.05
			{3.249	14.2}		
G	30	0.6254, 0.6270	{1.671	22.0}	1.47 ± 0.06	
	30		{1.598	13.7}		1.47 ± 0.03
	32	0.7105, 0.7117	1.882	6.4	1.48 ± 0.19	
F	27	0.8194, 0.8130	2.445	18.1	1.68 ± 0.07	1.67 ± 0.01
	28	1.032, 1.034	3.053	29.4	1.66 ± 0.02	
D	18	1.426, 1.428	{4.754	35.0}	1.87 ± 0.04	
	18		{4.734	23.0}		1.86 ± 0.01
	20	1.569, 1.569	5.207	31.5	1.86 ± 0.02	
C	13	1.013, 1.013	3.722	31.0	2.06 ± 0.02	2.05 ± 0.02
	16	0.9955, 0.9962	3.599	20.3	2.03 ± 0.07	
B	5	1.421, 1.417	5.137	25.7	2.03 ± 0.07	2.06 ± 0.03
	7	1.409, 1.409	5.252	29.7	2.09 ± 0.04	
A	1	0.9133, 0.9245	3.259	11.4	1.99 ± 0.08	2.00 ± 0.01
	2	0.9239, 0.9240	3.311	13.1	2.01 ± 0.02	
AA	10	0.8184, 0.8201	2.493	15.9	1.71 ± 0.04	(> 2.00)
	12	0.776, 0.772	2.795	15.0	2.44 ± 0.14	

* Radiogenic.
** Error limits represent estimate of one standard deviation for each individual run calculated in a manner similar to that described by Cox and Dalrymple [6].
*** Error limits represent one standard deviation calculated from all analyses made on each flow.

Table 2
Paleomagnetic data.

Flow	N	D	I	k	α_{95}	Demag. (Oe)	VGP Lat.	VGP Long.	Polarity
H	3	342°	− 63°	29	23°	150	77°	145°	N
G	4	197	66	53	13	NRM	− 79	190	R
F	3	337	− 55	103	12	150	68	230	N
D	4	186	72	84	10	NRM	− 82	260	R
C	4	354	− 59	86	10	150	78	263	N
B	4	16	12	8	35	150	32	307	I ?
A	4	165	68	49	13	600	− 81	10	R
AA	4	170	68	70	11	NRM	− 84	14	R

N = number of samples; D = mean declination, increasing eastward; I = inclination, positive downward; k = precision parameter [7]. α_{95} = semiangle of cone of 95% confidence [7]. Lat. = positive northern hemisphere; Long. = increasing eastward.

flows B and C are added, the resulting list of determinations would suggest that the Réunion events occurred from about 2.08 to 2.06 my and 2.00 to 1.96 my, because the boundaries would best be placed nearer the middle of the age overlaps of reversed and normal flows [10]. Similarly, the ages reported for flow H and a 1.01 my data reported by Chamalaun and McDougall [11] from Réunion Island suggest either that the Jaramillo event began about 1.01 my ago, rather than some 60 000 years later as suggested by Cox [8], or that this event is also split. Regardless of the exact timing, however, the record of all three major events in the Matuyama reversed polarity epoch in a single, continuous, radiometrically dated section provides confirmation of correlations made between deep-sea sediment cores or magnetic anomaly profiles and dated subaerial volcanic-rock sequences where some reversals are not represented.

Tills interbedded with the dated basalt flows (fig. 2) provide evidence of repeated glaciation of the site during the time spanned by the sequence. Mercer [2] has described the contact relations between the flows and tills, concluding that, although some of the basalt may have cooled in contact with ice, thick baked zones, including what are thought to be loess or soil horizons covering several tills, argue against such an origin. Petrographic evidence of subglacial origin was absent from all specimens studied, for most were holocrystalline and no hyaloclastic material was detected. Individual flows are traceable for as much as 4 km, suggesting that they flowed freely over a gently sloping surface. This evidence and the presence of fluviatile gravel and cobbles between flows A and B (fig. 2) indicate that the site was repeatedly ice covered and ice free between 2 my and 1 my ago.

The climatic implications of glaciation of this site 1 to 2 my ago are difficult to assess. Cerro del Fraile is in the rain shadow of the Cordillera and today has no permanent ice and only a few permanent snow beds on its eastern face. Nevertheless, the till lies only 100 to 200 m below the elevation of the firm limit on the ice field in the Cordillera to the west, from which outlet glaciers descend eastward to Lago Argentino at 186 m (fig. 1). In other words, if the landscape were less dissected, as it must have been 1 to 2 my ago, Cordilleran ice might cover the summit of Cerro del Fraile under present climatic conditions. The situation is complicated because the elevations of the Cordillera and Cerro del Fraile between 1 and 2 my ago are not known. Thus till deposition on Cerro del Fraile does not necessarily require that temperatures were lower then than they are today, but they are unlikely to have been much higher.

The earliest evidence for glaciation of Cerro del Fraile is the till underlying flow A-AA. As has been pointed out, the true age of this flow is uncertain, but considering its reversed polarity and the 2.06 my age of flow B, A-AA is probably between 2.08 and 2.43 my in age, if Cox's [8] value for the age of Gauss–Matuyama boundary is accepted. Deposition of cobbles shows that the site became ice free after the eruption of flow A-AA and prior to 2.06 my ago. Cerro del Fraile was again ice covered between 2.06 and 2.05 my ago. This thin till is the most precisely dated in the sequence. The area was again glaciated after 2.05 my and was ice free by 1.86 my ago; during this event, the thickest (12 m) and most laterally persistent till was deposited. It was glaciated twice between 1.86 and 1.67 my, and at least once between 1.47 and 1.03 my. After 1.03 my ago, Cordilleran ice did not again cover the hill. However, this does not imply decreasing intensity of glaciation, but rather reflects the increased dissection of the area that caused later incursions of ice to be confined to the glacially enlarged Lago Argentino depression.

It is difficult to compare these results with conclusions about southern hemisphere paleotemperatures inferred from the microfossil or coarse-sediment content of deep-sea cores or from marine strata exposed on land, because the workers involved have not agreed among themselves. Perhaps the closest parallel is with the results of Kennett et al. [12], who present a time scale for paleotemperature trends in a sequence of land-based marine sediments in New Zealand, using the data of Devereux et al. [13]. They use a reversal time scale consistent with that used here, but with different nomenclature for the events. They find the first major Pliocene cooling at 2.5 my ago, near the Gauss–Matuyama boundary; a cooling 2.13 to 2.10 my ago; a cooling 1.98 to 1.88 my ago; and one or perhaps two coolings 1.77 to 1.61 my ago. These authors, as well as others (*see*, for example, Berggren [14]), favor placing the Plio–Pleistocene boundary at about 1.8 my, that is, at or near the base of the Olduvai event. It should be noted, however, that this matter is still unresolved. Fillon [15] believed that

a major cooling occurred between the Réunion and Olduvai events (our terminology), that is, between 1.96 and 1.79 my ago. Bandy et al. [16] infer cooling from ca. 2.0 to 1.9 my ago, ca. 1.4 my ago, and ca. 1.2 my ago. They suggest a warmer climate between 1.8 and 1.6 my B.P. Hays [17] finds evidence for rapid cooling about 2.5 my ago and notes a sharp change in sedimentation in subantarctic deep-sea cores about 1.9 my ago, which he believes was caused by the start of large-scale freezing of the seas around Antarctica. Excepting the warming 1.8 to 1.6 my ago suggested by Bandy et al. [16], all of these alleged temperature fluctuations are consistent with the glacial record in southern Argentina.

5. Conclusions

Basalt flows at Cerro del Fraile, Argentina, dated by the potassium–argon method support evidence from other sources that at least three normal events occurred during the Matuyama reversed epoch. From presently available data, the Réunion event of Grommé and Hay [9] includes multiple reversals, with one normal event at 2.05 ± 0.05 my. Ages of normally magnetized basalts of 1.03 ± 0.05 my in Argentina and 1.01 ± 0.02 my on Réunion Island suggest that either the Jaramillo event began somewhat earlier than the 0.92 to 0.95 my age assigned by a number of workers or that it is also a split event. Glaciers in southern Argentina extended to Cerro del Fraile in the Andean foothills at least five times before 1.67 my ago, with the first incursion probably occurring before 2.08 my B.P. At least one incursion occurred between 1.47 and 1.03 my, but no indication was found that the site was ice covered between 1.67 and 1.47 my B.P.

Acknowledgements

Potassium–argon ages reported here were determined at The Ohio State University with support provided by the Department of Geology. Field work in southern Argentina was funded by National Science Foundation Grant GA-881 and by the Institute of Polar Studies, The Ohio State University. Dr. Walter Sander of the Direccíon Nacional de Agua y Energía, Buenos Aires, assisted in the collection of the oriented samples. Magnetic measurements were made in the Department of Geology, Case Western Reserve University, which provided support for that work. The authors would also like to thank G.B. Dalrymple, R.J. Janda, and C.S. Grommé for their critical reading of the manuscript.

References

[1] E. Feruglio, Estudios geológicos y glaciológicos en la région del Lago Argentino (Patagonia), Bol. Acad. National Ciencias Córdoba 37 (1944) 3.
[2] J.H. Mercer, Glaciation in southern Argentina more than two million years ago, Science 164 (1969) 823.
[3] J.A. Cooper, The flame photometric determination of potassium in geological materials used for potassium–argon dating, Geochim. Cosmochim. Acta 27 (1963) 525.
[4] J.F. Evernden and G.H. Curtis, The potassium–argon dating of late Cenozoic rocks in East Africa and Italy, Current Anthropology 6 (1965) 343.
[5] J.H. Foster, A paleomagnetic spinner magnetometer using a fluxgate gradiometer, Earth Planet. Sci. Letters 1 (1966) 463.
[6] A. Cox and G.B. Dalrymple, Statistical analysis of geomagnetic reversal data and the precision of potassium–argon dating, J. Geophys. Res. 72 (1967) 2603.
[7] R.A. Fischer, Dispersion on a sphere, Proc. Roy. Soc. Lond., ser. A 217 (1953) 295.
[8] A. Cox, Geomagnetic reversals, Science 163 (1969) 237.
[9] C.S. Grommé and R.L. Hay, Geomagnetic polarity epochs: Age and duration of the Olduvai normal polarity event, Earth Planet. Sci. Letters 10 (1971) 179.
[10] G.B. Dalrymple, Potassium–argon dating of geomagnetic reversals and North American glaciations, *in* Calibration of Hominoid Evolution, W.W. Bishop and J.A. Miller, eds. (Scottish Academic Press, Edinburgh) in press.
[11] F.H. Chamalaun and I. McDougall, Dating geomagnetic polarity epochs in Réunion, Nature 210 (1966) 1212.
[12] J.P. Kennett, N.D. Watkins and P. Vella, Paleoclimatic chronology of Pliocene-Early pleistocene climates and the Plio-Pleistocene boundary in New Zealand, Science 171 (1971) 276.
[13] I. Devereux, C.H. Hendy and P. Vella, Pliocene and early Pleistocene sea temperature fluctuations, Mangaopari Stream, New Zealand, Earth Planet. Sci. Letters 8 (1970) 163.
[14] W.A. Berggren, A Cenozoic time-scale–Some implications for regional geology and paleobiogeography, Lethaia 5 (1972) 195.
[15] R.H. Fillon, Paleoclimatic fluctuations in glacial-marine sediments of Antarctic deep-sea cores, Geol. Soc. Am. Abstracts with Programs 2, no. 7 (1970) 550.
[16] O.L. Bandy, R.E. Casey and R.C. Wright, Late Neogene planktonic zonation, magnetic reversals, and radiometric

dates, Antarctic to the tropics, *in* J.L. Reid, ed., 1971, Antarctic Oceanology I, p. 1-26, Am. Geophys. Union Antarctic Res. Ser. 15 (1971) p. 343.

[17] J.D. Hays, Climatic record of late Cenozoic Antarctic Ocean sediments related to the record of world climate, *in* E.M. van Zinderen Bakker, ed., 1968, Palaeoecology of Africa and of the surrounding islands and Antarctica 5, p. 139-163 (Balkema Publishers, Cape Town, South Africa, 1968) p. 240.